

PixelPlayers: Automated Whiteboard Basketball Annotations

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Abstract

*Basketball coaches and players rely heavily on game film to review opposing formations and develop counter-strategies, a process that typically involves manually scrubbing footage and sketching out plays on a whiteboard. While this workflow is central to how teams prepare, it can be time-intensive across a full season of film. Our goal is to augment this process by generating visual patterns or formations for coaches and players to work with. We present **PixelPlayers**, an end-to-end framework that derives basketball player and ball tracking instances through fine-tuned video supervision. The system produces annotated whiteboard-style play diagrams overlaid on footage, designed to serve as a starting point for game analysis.*

1. Introduction

Basketball teams around the world invest significant time in reviewing game footage to identify common opponent formations, and to develop counter-strategies. In practice, this process relies heavily on coaches and players manually scrubbing game footage, and hand-drawing player movements around the court on a whiteboard — a workflow that is intuitive but difficult to scale. Bridging this gap in a generative fashion in a way that would be useful for play preparation is non-trivial: for one, sports scene understanding is inherently noisy, as many key objects are frequently occluded and subject to motion blur as the stadium cameras pan to track players.

To address this limitation, we present **PixelPlayers**, an end-to-end automated annotation pipeline designed for basketball game footage. Our framework produces per-frame player and ball detections via fine-tuned video supervision and a planar image-to-court homography estimate from auto-inferred court quadrilaterals, allowing rendering of player and ball trajectories in canonical court coordinates in the form of a whiteboard-style play diagram. We hope that this foundation will better lend support to critical downstream tasks in sport analysis, including but not limited to play segmentation and classification, and pattern retrieval across large video corpora.

2. Related Work

Automated sports video analysis has been an active area of research, with prior work spanning player detection and court localization. Early approaches to player detection in video footage relied on background subtraction and color histograms, but modern pipelines have largely converged on deep-learned feature detectors such as Faster R-CNN [3] and the YOLO family [2] for their speed-accuracy trade-off in real-time settings. In the field of basketball specifically, the DeepSportRadar dataset [4] provides a standard way to calibrate imagery from various different basketball games, providing player and ball instance annotations.

While multi-object tracking in sports video has been approached through dedicated learned trackers and simpler tracking-by-detection schemes, with SORT [1] and its appearance-aware extension DeepSORT [5] remaining widely used baselines, end-to-end sports analytics systems combine many of these components at the cost of reproducibility and interpretability, and usefulness for coaches and players of the games. **PixelPlayers** draws on this body of literature to assemble a lightweight framework tailored to basketball footage review.

3. Method

PixelPlayers' pipeline is split into two parts: (1) YOLO-based player and ball detection and localization, (2) DINO-based court line detection with geometric homography estimation, and an output visualizer head at the end.

3.1. Detection and localization

We export DeepSport dataset image supervision into YOLO labels using two complementary annotation sources. First, when a per-frame instance mask is readily available, we recover object instances directly from pixel values, and convert them into bounding boxes. Second, we use index annotations together with per-frame camera calibration to project 3D metadata into the image, and derive bounding boxes from projected keypoints. Boxes are then converted to YOLO's normalized input format (c, x, y, w, h) and we split training and validation tests sets by game ID.

When segmentation masks are available, we treat each dis-

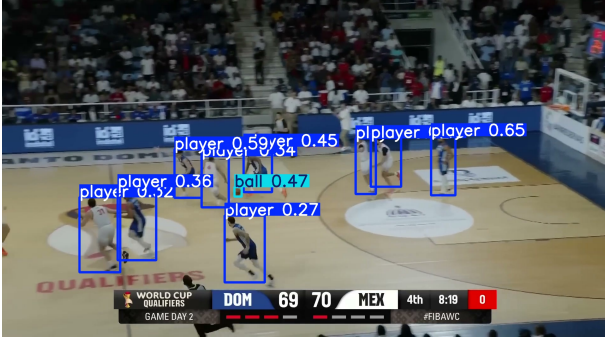


Figure 1: Player and ball detection instance on a broadcast footage frame.

tinct non-background mask value as one object instance, and wrap each with the smallest possible bounding box. When masks are not available (or do not contain a given class), we instead use the DeepSport ground truth annotations in addition to the provided camera calibration to project labeled 3D information into the image. For the basketball itself, we project its annotated center and place a small square box around that point. For players, we project body keypoints instead. Finally, we decide which supervision to use separately for each class. For example, if mask-derived boxes exist for players, we use them. Otherwise, we fall back to the calibration-projected boxes for that class. As a result, a frame may mix supervision sources for most stable training.

We then fine-tune YOLOv8 using Ultralytics. At inference time, we convert per-frame detections into short track IDs using a minimal centroid tracker: within each class, each detection is greedily matched to the nearest unassigned active track ID in image space if its center is within a pixel distance threshold. Unmatched detections start new tracks; unmatched tracks increment a `stale` counter and are removed once `stale` exceeds a fixed threshold. This matching step costs $O(\sum_c N_c M_c)$ per frame and is easily interpretable, but can swap IDs or fragment tracks under occlusions and crossing motion, which we further analyze in Results.

3.2. Court line detection and homography estimation

Our court localization pipeline separates visual detection from geometric estimation. First, a dense neural network predicts image-space evidence for FIBA court markings: a binary line heatmap, a marking-class distribution, a court mask, and a left/right court-side map. These predictions convert the raw broadcast image into a structured intermediate representation that is easier to match against the known court template. Second, we fit geometric primitives to the predicted marking evidence and estimate a planar homography from court coordinates, measured in centimeters, to image pixels.

3.2.1 Line Detection Network

The line detector is a U-Net-style dense prediction model with a frozen DINOv3 ConvNeXt backbone. Multi-scale backbone features are projected into a shared decoder for local court-marking prediction, while a separate lightweight layout decoder uses deeper features for global court structure. We output the following channels:

1. Dense "lineness" heatmap
2. Dense "line class" feature map, using the following classes: far sideline, near sideline, side sideline, three point arc, free-throw circle, far foul line, and near foul line.
3. "Sidedness" map, parameterized by a single line in image space. Note that line classes are shared between the left and right side of the court, while the sidedness map adds the court side information.
4. "Courtiness" mask.

Figure 2 visualizes these channels. Training targets are rendered directly from the DeepSport camera calibration and a metric FIBA court model. Each physical marking is sampled in world coordinates, and projected into the image. The lineness head is trained with a Dice and focal binary loss, the class head with cross entropy gated by the line target, the court-mask head with binary cross entropy, and the side head with a court-mask-weighted binary cross entropy.

3.2.2 Homography Linear DLT with RANSAC

Given the network outputs, we form class-specific evidence maps by multiplying lineness and class probability, weighted by the predicted court mask (Figure 2f). Weighted primitive fitting produces candidate image lines for straight markings and ellipse fits for curved markings. Shared semantic classes are split back into side-specific geometric names using the court-side prediction (Figure 2d), so the estimator can reason separately about, for example, the left and right baselines even though they share a network class.

Each homography candidate is estimated with a normalized linear DLT system using both line and point correspondences, each contributing 2 linear equations:

$$\begin{aligned}\ell_i &\sim H^{-T} \ell_w, \\ \mathbf{x}_i &\sim H \mathbf{x}_w.\end{aligned}$$

The DLT system is solved by SVD after Hartley normalization of world and image coordinates. In addition to lines and corners, additional point constraints are derived from curved primitives (three-point arc and free-throw circle). Curve-derived point constraints are upweighted because arcs and circles are fitted more reliably even on noisy frames.

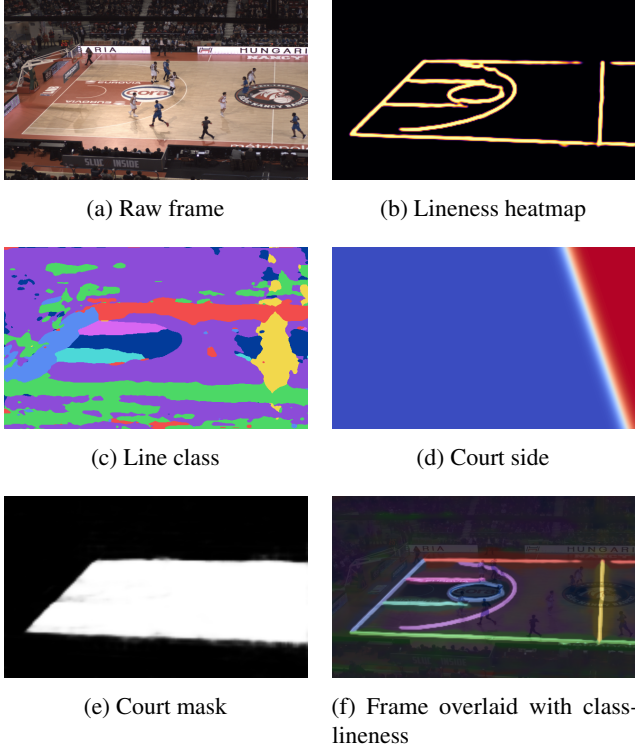


Figure 2: Example output from the court line detection network

To handle ill-fitted and missing markings, candidate homographies are generated in a RANSAC-style loop over plausible primitive subsets. We enumerate the small combinatorial space of top primitive candidates while allowing up to two primitives to be excluded. After an initial DLT solve, the system rejects geometrically degenerate projections. Candidates are then scored by projecting the full sampled FIBA marking template into the image and reading network evidence near each projected sample. The score is length-weighted in world space and multiplied by predicted court-mask and court-side probabilities when applicable, so unsupported long markings are penalized more than short or ambiguous structures. The selected homography is therefore the one that best explains the complete visible court layout, not merely the one that fits a minimal subset of detected primitives.

3.2.3 Data augmentation

In order to imitate real footage condition, we introduced an aggressive data augmentation scheme, applying these transforms at random: (1) cropping, (2) color shift, (3) overlay rectangle (imitating score bar in a real footage), and (4) virtual camera panning by stitching frames from multiple viewpoints to artificially create varying camera views.



Figure 3: Geometric primitive fitting and homography estimation via RANSAC. Despite an incorrect fit on the left near foul line (cyan), the overall homography estimate remains robust.



Figure 4: Left: a real unannotated broadcast frame. Right: three examples of augmented DeepSport frames, showing cropping, color shift, score bar overlay, and virtual camera panning.

4. Results

4.1. System Outputs

Our evaluation focuses on qualitative robustness and end-to-end system behavior under realistic basketball broadcast conditions, rather than quantitative benchmarks. This choice is intentional, as **PixelPlayers** is designed to be a full

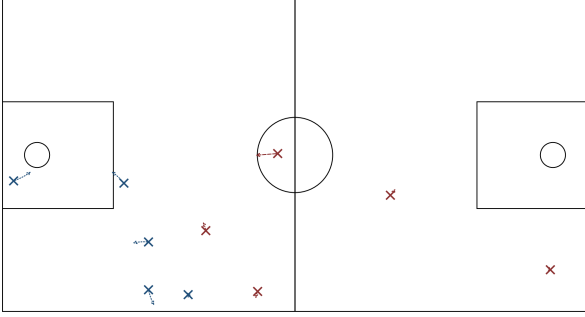


Figure 5: Whiteboard-style coaching diagram generated from a short video sequence.

pipeline for generating interpretable coaching style visualizations, where failure modes are often structural, such as broken trajectories or misaligned court geometry, and better captured through visual inspection rather than hard metrics.

Overall, the system produces: **(i)** temporally consistent player tracks for most visible players, **(ii)** a ball track when the ball is reliably detected, and **(iii)** plausible court-coordinate trajectories when court geometry is correctly estimated via calibration or stable auto-inferred quadrilaterals. Despite operating on a limited set of annotated frames (approximately 60 frames per sequence), the pipeline produces coherent whiteboard-style visualizations that preserve tactical structure and provide an interpretable summary of player movement across the court over time.

Notably, both court homography estimation and player detection yield strong qualitative alignment even under challenging broadcast conditions. Court geometry is recovered reliably enough to support meaningful projection into canonical court space, while player detections remain sufficiently stable to support coherent short-term trajectory reconstruction in the 2D image.

4.2. Evaluation

We define success as producing an annotated video where boxes and IDs are stable for players in view, a field plot where trajectories lie in court bounds and match the direction of motion in the video, and a whiteboard diagram that is readable for coaches. These criteria are aligned with the intended stakeholder product as a visual analytics tool (a coaching sketch) rather than a fully calibrated measurement system, which would not be helpful for coaches and players seeking to develop strategies off of our tool.

4.3. Failure modes

Ball detection. The ball is often only a few pixels wide and frequently motion-blurred, so missed detections lead to fragmented or missing ball trajectories. Bright circular logos and highlights can also trigger false positive detections of

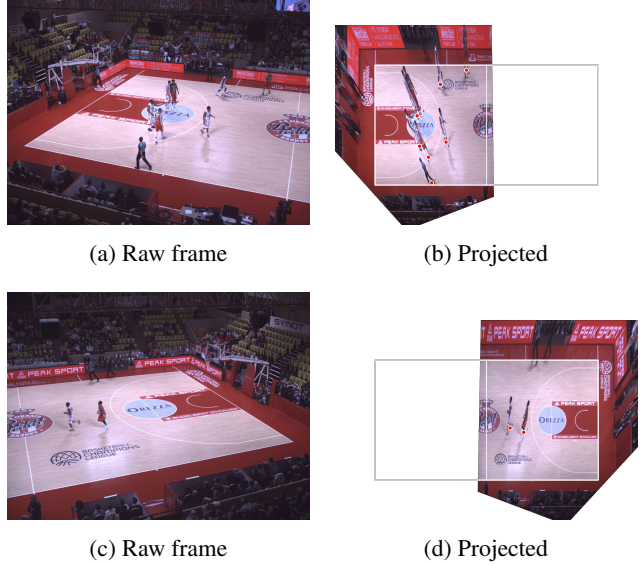


Figure 6: Example output reprojected into canonical court coordinate using homography

the ball, which can be seen in different basketball arenas.

ID switches. Centroid tracking specifically is unstable under occlusion and player crossings, which can cause identity swaps and broken trajectories in generating the final whiteboard visualizations. This failure mode is most blatant in longer videos.

Domain gap. DeepSportRadar [4] contains clean broadcast-style data, while real footage often motion blur, lighting variation, camera panning, and compression artifacts. Score overlays often occlude key court lines, making homography estimation less reliable. Although we use augmentation to simulate these effects, generalization to typical game footage remains limited.

Summary. Overall, while individual failure modes are expected, the system still maintains strong end-to-end visual coherence, demonstrating that meaningful player motion and court level structure can still be recovered even from a small number of video frames from real game-play.

5. Technical Discussion and SRC Response

5.1. Impact on Coaches and Players

Although this system is a coaching aid, it produces artifacts that can easily be repurposed for scouting or even player evaluation, potentially without players' awareness. This raises important concerns about how automated outputs may be interpreted with respect to other heuristics in the league.

Furthermore, teams may choose to use the data to inform decisions on rosters, and leagues may adopt it for analytics-driven marketing. In any of these cases, the precision of the

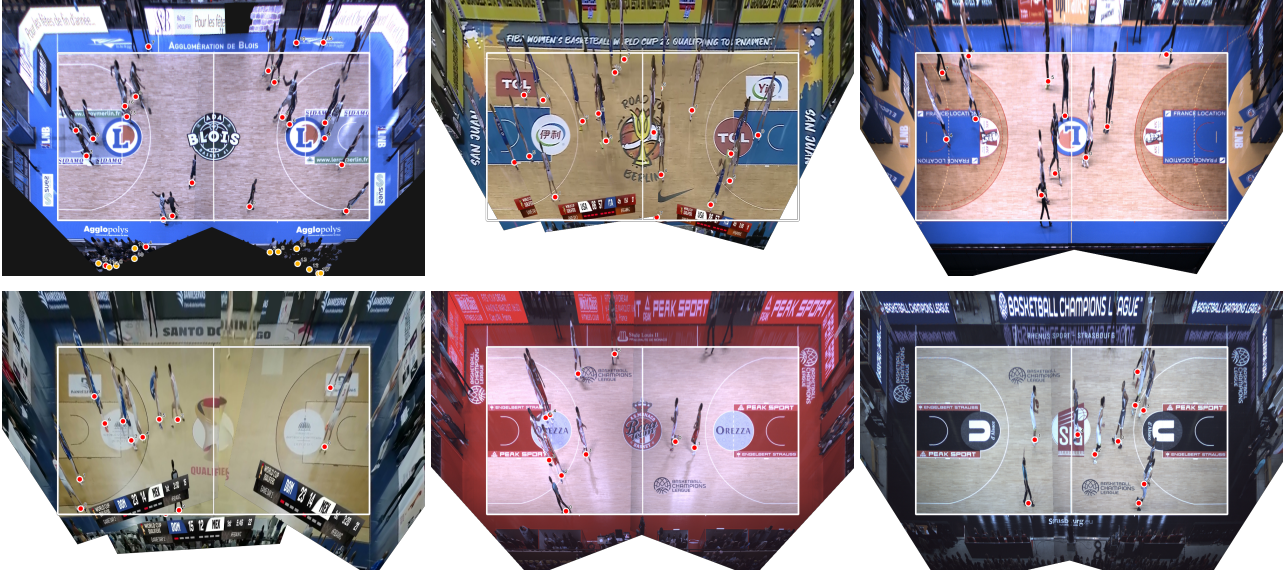


Figure 7: Full-court visualizations produced by overlaying homography-based reprojections from multiple camera angles. Score bar overlays from real broadcast footage appear as visual artifacts in some frames.

output is not only of utmost importance, but can also hide underlying model assumptions. This makes it important to frame the system’s outputs as interpretive artifacts rather than definitive representations of on games on the court.

5.2. Role of the human critic

By reducing gameplay to spatial trajectories, the system can tend to omit intent, timing, player quirks, and tactical decision-making that human analysts infer from broader game context. Elements such as off-ball spacing, play design, and situational awareness are not recoverable from just bounding boxes. We argue that human interpretation plays a critical role in understanding these nuances, and basketball analytics has already shown a tendency to shift play styles toward what is most measurable. A system that further lowers the cost of positional analysis risks reinforcing this trend if its outputs are treated as authoritative rather than as one component within a broader analytical framework. Overreliance on our artifacts may narrow how performance is understood, whereas we intend the system to be used to supplement coaching and game-play analysis, not hinder it.

5.3. Data bias

The training data is derived primarily from professional European basketball broadcasts, which is not fully representative of other leagues or player demographics. This introduces distributional bias that affects both detection and tracking performance and across diverse uniforms and body types.

These errors are also not uniformly distributed across players or positions. In particular, interior players such as

centers and power forwards are more likely to be affected due to frequent congestion near the paint, which can lead to systematic under-tracking of their contributions relative to guards operating in more open space.

5.4. Other applicability concerns

This system is designed for basketball under FIBA court geometry (28×15 m), with fixed assumptions about court markings, object classes, and the structure of a whiteboard play diagram. While the underlying components, such as YOLO-based detection and homography estimation are general-purpose, their integration is coupled to basketball-specific rules and structure.

Extending the framework to other basketball associations, such as the NBA or WNBA, or to other sports would require redefining geometric priors and annotations. Such extensions would therefore need to be treated as full domain adaptation problems, rather than just transferring our current pipeline.

6. Conclusion

In basketball, a common way to improve one’s game-play and improve one’s strategies is to review game footage. In practice however, this process is time-consuming and difficult to scale across the thousands of hours players spend on the court, as well as the need for generalization over an abundance of games, which may be difficult to keep in context for most players and coaches.

To address this issue, we built an end-to-end prototype for automatic basketball play annotation using YOLO detection fine-tuned on DeepSport data, centroid tracking, and

homography projection to a canonical court. Our framework produces the artifact of a whiteboard flow-sketch, intended to be of use to both coaches and players who seek to better understand their own plays against opponent formations.

6.1. Future work

The highest-impact improvements to our framework would be adding appearance and motion models for more robust tracking and learning court layout detection with temporal smoothing to supplement our current heuristics. We would also encourage this framework to be applied to other sports with the need for whiteboard-like strategy making, where such a framework could be useful to supplement tactical understanding, and ultimately lead to more iterations over in-game decisions.

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References

- [1] Alex Bewley, Zongyuan Ge, Lionel Ott, Fabio Ramos, and Ben Upcroft. Simple online and realtime tracking. In *IEEE International Conference on Image Processing*, 2016.
- [2] Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics YOLOv8, 2023. <https://github.com/ultralytics/ultralytics>.
- [3] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems*, 2015.
- [4] Gabriel Van Zandycke, Vladimir Somers, Maxime Istasse, Carlo Del Don, and Davide Zambrano. DeepSportradar-v1: Computer vision dataset for sports understanding with high quality annotations. In *Proceedings of the 5th International ACM Workshop on Multimedia Content Analysis in Sports*. ACM, 2022.
- [5] Nicolai Wojke, Alex Bewley, and Dietrich Paulus. Simple online and realtime tracking with a deep association metric. In *IEEE International Conference on Image Processing*, 2017.

Appendix

Team contributions

Oras Phongpanangam Designed and implemented the court line detection pipeline, including geometric primitive fitting, RANSAC-based DLT homography estimation, and data augmentation for robust court structure

inference under broadcast conditions. Evaluated homography stability across varied camera views.

Jeffrey Mu Designed and implemented YOLOv8-based player and ball detection, including dataset construction from DeepSport annotations, label conversion pipelines, and augmentation strategies for broadcast domain adaptation. Was responsible for fine-tuning the model, integrating with Oras’ homography analysis, and contributed to work on producing the final whiteboard flow visualization.

Arib Syed Designed and implemented the end-to-end spatial representation and system integration pipeline, connecting detection, tracking, and homography estimation into a unified coordinate framework. Developed the full visualization and projection system for field plots, annotated video, and whiteboard-style tactical reconstruction, ensuring temporal and spatial consistency across modules. Contributed to cross-module debugging, particularly addressing inconsistencies between detection drift and tracking stability.

Heewon Seo Defined validation and evaluation protocols and implemented temporal stabilization strategies for multi-object tracking, including centroid-based association, trajectory smoothing, and consistency metrics. Tuned tracking and detection thresholds to improve robustness in crowded and occluded scenes, and evaluated system stability under broadcast-like conditions.

Output assets

In addition to the visualized frames, we also applied our pipeline to two video clips of real broadcast footage. There are available on Google Drive, and are also included in the repo. (1): https://drive.google.com/file/d/1ZEVxonLmpMUow6xzriIkEDg6M6lPiU1C/view?usp=drive_link, (2): https://drive.google.com/file/d/16mKEYOc05QkJy3sljUrbiv3McboDulaY/view?usp=drive_link

Model checkpoints are too large to push to GitHub and are thus hosted on Google Drive: https://drive.google.com/file/d/10FG0r94FJN4_N4rs45bJI-umOGt_8N88/view?usp=sharing

All other results can be found in the GitHub repo itself.